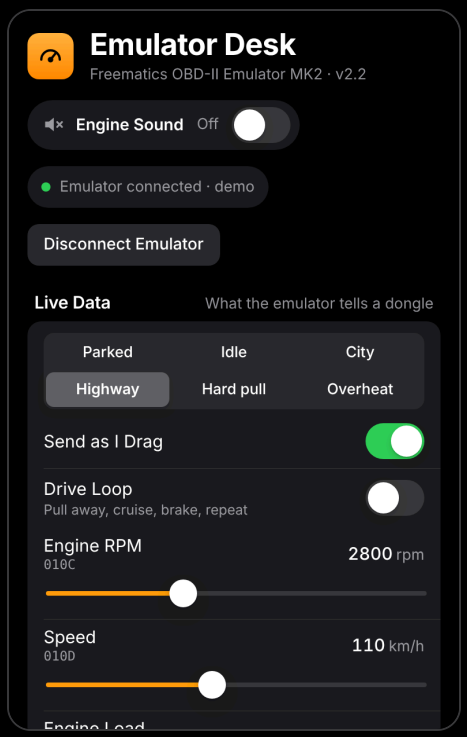
# Emulator Desk

A Mac control panel for the Freematics OBD-II Emulator MK2

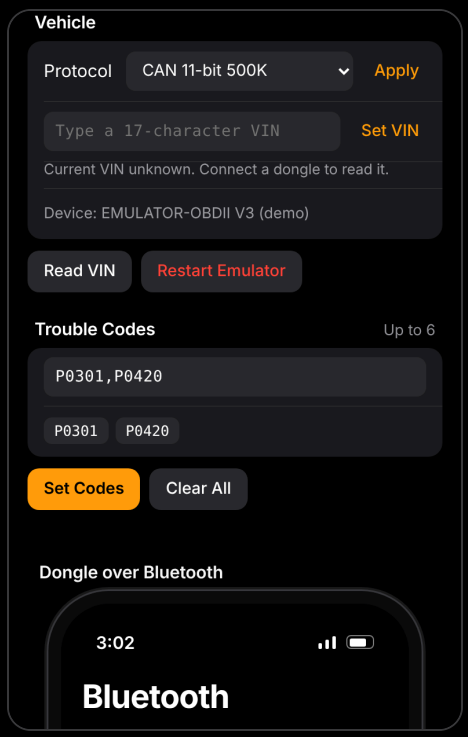
App **v2.2**
Document **v2.2**

Emulator Desk drives a Freematics MK2 emulator from Google Chrome over its USB cable, so a Mac can do what the Windows-only Freematics GUI does. You set live engine values, the VIN, the bus protocol and trouble codes, and any OBD-II dongle or scan tool plugged into the emulator reads them as if it were in a real car. It runs from a web page, with nothing to install, and follows Apple's Human Interface Guidelines in light and dark mode.



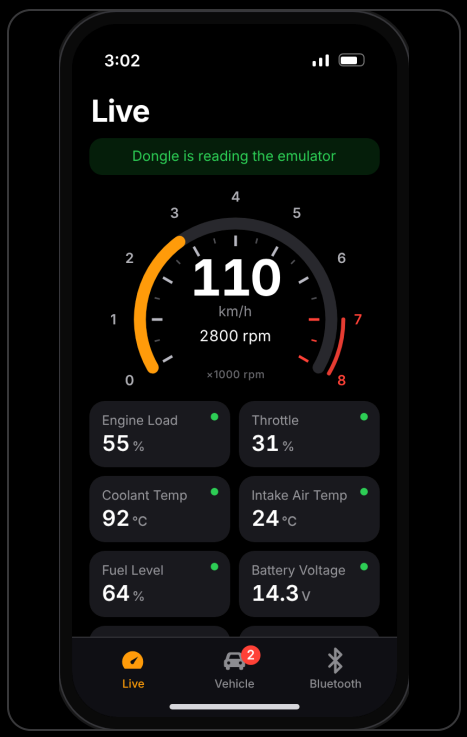

### Live data

Ten sliders for the common Mode 01 values, with presets from Parked to Overheat. Values go out as you drag.




### Vehicle and codes

A compact panel for protocol, VIN and up to six trouble codes, set or cleared in one tap.




### Dongle phone view

A phone with a tab bar. Live has an rpm gauge with redline, Vehicle decodes the VIN, Bluetooth shows the link and what the dongle supports.

| 10 | 6 | 7 | 1 |
|---|---|---|---|
| live PIDs with sliders | driving presets plus a drive loop | bus protocols to choose from | link to open it, nothing to install |

- **On your phone**

phone.html is the same Live, Vehicle and Bluetooth app, full screen. Scan the QR code under the phone; on iPhone open it in Bluefy.

- **Dongle over Bluetooth**

Connect a Bluetooth LE ELM327 dongle from the Bluetooth tab. Dots show matches, and Vehicle decodes the VIN it reads into make, model, engine and origin.

- **Good to know**

Sliders read back the emulator's values on connect. Its firmware can't report its VIN, so the VIN box fills in from the dongle.

- **Engine sound**

Flip the Engine Sound switch and a flat-six plays, built from single cylinder firings. It thumps at idle, howls near redline and crackles on lift-off.

# Technical setup

Sources, integrations and what runs where

App **v2.2**
Document **v2.2**

## Sources and libraries

| Piece | Version | What it does | Loaded from | Licence |
|---|---|---|---|---|
| Emulator Desk | 2.2 | The whole app: UI, command queue, PID encoding, presets, drive loop, engine sound, dongle check | `index.html` and `phone.html` on GitHub Pages | Written for this app |
| Freematics AT command set | EMULATOR-OBDII V3 (tested) | Commands the app sends: `ATE0`, `ATZ`, `ATSET`, `ATGET` (PIDs), `ATCLR DTC` | Built into the emulator | Freematics |
| OBD-II PID formulas | SAE J1979 | Byte encodings for each slider (e.g. RPM × 4 as two bytes, temperature + 40) | Hard-coded in the app | Public standard |

No third-party libraries, CDNs or web fonts are used; type is the system font.

## Integrations

| Integration | Used for | Details |
|---|---|---|
| Web Serial API | Talking to the emulator | `navigator.serial.requestPort()` filtered to CP2102 and CH340 USB chips so a paired dongle can't be picked, opened at 38400 baud, 8N1. You pick the port in Chrome's chooser; no auth. |
| Freematics OBD-II Emulator MK2 | The device being controlled | USB to serial through a CP2102 or CH340 bridge, both with drivers built into macOS. Needs its 12V adapter for power. |
| Web Bluetooth | Reading a plugged-in dongle (Mac or phone) | `navigator.bluetooth.requestDevice()`, you pick the dongle. Finds a notify and a write characteristic (FFE0, FFF0, 18F0 and similar), sends ELM327 commands: setup (`ATZ`, `ATSP0`), capabilities (`AT@1`, `STI`, `ATDP`, `0100–0160`, `011C`, `0900`) and live polling. Writes in 20-byte chunks, times each reply and keeps a copyable connection log. |
| NHTSA vPIC | VIN details | `vpic.nhtsa.dot.gov/api/vehicles/DecodeVinValues/{VIN}?format=json`, no key. Sends only the VIN; returns model, engine, body and plant. Maker, country, year and check digit decode offline. |
| Web Audio API | Engine sound | An AudioWorklet fires six cylinder pulses every two revolutions through exhaust resonators, a muffler filter, EQ and a compressor. Starts only when you flip the switch. |
| GitHub Pages | Hosting | `davidanthonyalleyne-dot.github.io/emulator-desk`, served from the `main` branch of the public `emulator-desk` repo. Static files only. |

## What runs where

**On your Mac:** everything. The page runs in Chrome and sends one command at a time, waiting up to 1.2 seconds for OK or ERROR before the next.

**Network:** the page load from GitHub Pages, plus one NHTSA lookup per new VIN decoded.

**Data kept:** nothing. Slider values, VIN and codes live only in the open page and on the emulator itself.

**Demo mode:** add `?demo` to the address to try the app without hardware. The screenshots in this document use it with sample values.

## Regenerating this document

In `docs/overview`, run `node shoot.js` to retake the screenshots in demo mode at 390 × 844, then `node pdf.js` to print `overview.pdf` on Letter pages.